Metal Detecting: A Complete Masterclass from Beginner to Expert

Kevin Hoagland and Gold Prospectors Association of America

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Introduction: It's Not Complicated — And I'm Living Proof

My name is Kevin Hoagland. I'm the Executive Director of Development for the Gold Prospectors Association of America and host of *Gold Trails*. I've been metal detecting a long time — a very long time. I built my first detector when I was nine years old, back in 1970, and I've been fascinated with everything about this hobby ever since.

That little advanced-level kit machine wasn't fancy. And I learned a great deal when I had it completely built and would find my mother's from a foot away. I knew it could do better, and I began my journey to make it just that: better. I spent hours at my local Radio Shack and learned everything I could from the staff. I spent so much time there that they gave me a name badge. I also spent countless hours at the library. Both paid off. It didn't have a dozen knobs, a digital screen, or Bluetooth. It just beeped when it found something, and when I was done, I could find her car from five feet away.

That was enough to hook me for life. Fifty-five years later I still get the same thrill every time I hear a good signal break through the threshold. I've hunted nuggets in the desert, relics in old ghost towns, rings on beaches, and coins in parks all across this country and across the world. I've taught thousands of people in GPAA seminars, at LDMA claims, and right here in this masterclass. And you know what I tell every single one of them?

Metal detecting is not complicated.

It may *feel* complicated when you unbox that shiny new machine and stare at all the buttons, dials, and menu options. I get it. I've watched grown men and women stand in the middle of a claim looking like they just opened a nuclear reactor manual. But here's the truth I've lived every single day since I was a kid: once you understand the five simple things

that *every* detector does the same way — no matter if it's a $200 starter or a $10,000 flagship — the whole thing becomes dead simple.

That's why I'm writing this book.

I want you — whether you're holding your very first detector today, or you've been swinging one for twenty years and still feel like you're missing half the targets — to finally get it. Really get it. Not the tech-speak version. Not the "buy my expensive coil and you'll find more gold" sales pitch. The real, field-tested fundamentals that work on *any* machine, in *any* ground, for *any* kind of treasure.

In the chapters ahead I'm going to show you exactly what I teach in my live classes and what I've used to find gold that other detectorists walked right over. We're going to talk ripples in a pond, not electromagnetic theory. We're going to set your threshold so it sounds like a mosquito in your ear instead of a jet engine. We're going to ground balance like we're shaking hands with the dirt instead of fighting it. And we're going to grid a patch so systematically you'll never wonder again if you "just missed it."

This isn't a book about one brand. It's not about chasing the latest gadget. It's about the similarities — the core DNA that every detector shares — so you can pick up *any* machine tomorrow and hunt with confidence.

Here's my promise to you:

By the time you finish this masterclass, you will understand your detector better than 90% of the people swinging one right now. You'll know exactly why it does what it does. You'll stop guessing and start *finding*. And most importantly, you'll stop feeling overwhelmed and start having fun again — the same kind of fun I had when I was nine years old and the whole world was one big ripple waiting to come back to me.

Please think about this every time you go out detecting; this is the first thing I want to get into your head.

"Your detector can do one thing that no human being has, can or will ever do, and that is to see one billionth of an inch under the ground. When it talks to me, It is my job to understand what it sees."

How to use this book? Simple.

Read it once straight through like we're sitting on a tailgate after a long day on the claim. Then take it with you — dog-ear the pages, highlight the drills, snap photos of the checklists. Treat it like your field manual. Because that's exactly what it is.

The gold, the relics, the rings, the coins — they're all still out there. They've been waiting.

And now you've got the map.

Let's go find them.

— Kevin Hoagland

Executive Director of Development, GPAA/LDMA

Host, *Gold Trails*

Chapter 1: Answering the Three Questions I Get Asked Everywhere

I've been doing this for fifty-five years, and no matter where I am — GPAA seminar, LDMA claim, YouTube comments, or standing in the middle of a desert wash — I get the same three questions over and over.

Every. Single. Time.

So let's knock them out right now, straight and honest, the way I do in every class. No sales pitch. No "buy this machine and you'll be rich." Just the real-world truth from someone who's actually been out there swinging.

"How deep will it go?" — The honest answer every beginner needs

There is no number. There is no "this machine goes 12 inches" guarantee.

I've had people hold up a nice little nugget and say, "I've got this gold nugget right here and it's buried six inches — should be able to find that with any detector, right?"

In perfect ground? Maybe.

Now let me put a solid cap of basalt over it. Or a layer of iron magnetite. Or hot red mineralized dirt like we have in half of Arizona. Suddenly that same nugget might as well be on the moon.

Depth is 100% about the ground you're hunting in. Mineralization changes everything. The machine has to "see" through the earth before it can see your target. That's why ground balancing (we'll cover it deep in Chapter 6) is the single most important skill you'll learn.

The honest answer? It goes as deep as the ground and your settings will let it. Test it yourself on every new patch. Always carry a confidence (aka test piece) target with you. I carry a small gold nugget glued to a poker chip. I'll bury it if the ground iron is not too bad and just set it on the top in really bad magnetic ground. Then swing over it and pay attention to the response. It may sound very clear or it could sound a little muddy. If muddy, you have a little bit of fine tuning to take care of.

"How small of a gold nugget can it find?" — Why that vial test fails and what really matters

This is the number two question I get, and it's the one that causes the most frustration.

A guy takes a little vial full of fine gold flakes, sets it on the ground, swings his coil over it... and nothing. Dead silence. He thinks his detector is broken. It's not.

Here's why: your detector sends out a signal that creates eddy currents in a target. That target has to be large enough — singular piece large enough — to generate a return signal that makes it back to the coil. A thousand microscopic flakes all trying to talk at once just interfere with each other. The machine gets confused and stays quiet.

One solid little picker the size of a match head? That can scream. A vial of flour gold? Often invisible.

High-frequency machines help with tiny gold, sure. But even then, mineralization of the ground is the boss. In mild ground I've seen tiny pieces light up. In nasty hot ground, the same piece disappears.

Bottom line: there is no universal "smallest gold it can find." Too many variables. The real test is single pieces in the actual ground you hunt — not a vial on your kitchen table.

"Is there a true gold-only detector?" — The vacuum-of-space joke + why pure gold doesn't exist here on Earth

Nope. Not a one. And if there was, you wouldn't be reading this written by me because I would have a gold-only detector and would be in the field every day.

And here's why: there is no such thing as pure gold in nature on this planet. None. Zip.

We all know gold does not have a particular single frequency response. Real-world gold is alloyed with silver, copper, iron or whatever else Mother Nature threw in.

If you had absolutely pure, 99.999 fine gold... and you had the luxury of hunting with your detector in the vacuum of space... then yeah, maybe we could make a true gold-only machine work.

So if you ever get a chance to go on the space shuttle, take your detector on a spacewalk, and swing over some pure gold out there in the void — please get back to me and let me know how that worked out. Because here on planet Earth, where the rest of us are hunting, it doesn't exist.

We hunt gold that's mixed with everything else, sitting in dirt that's throwing its own signals back at us.

The Mineralization Reality Check That Changes Everything

What is the largest target you will ever detect with your detector?

The earth.

The ground itself is one massive target. Hot ground, hot rocks, black sand, magnetite, basalt, salty beach and then there are cold stones — all of it is screaming back at your coil every swing. Your job is to make the ground "disappear" so the real treasures can stand out.

That's why factory presets are great for starting, but real hunting means learning your machine in the actual dirt you're standing on.

One more real-world tip I give every class: turn your cell phone off. Or at least put it in airplane mode. I've watched prototype machines go crazy because someone's phone was in their pocket two feet away. Interference is real, and it kills depth and small-target sensitivity faster than anything. This is called EMI Electromagnetic Interference and in the world today, it is everywhere. Aircraft, cell towers, wireless internet towers, even Starlink Satellites can cause EMI.

Action Steps — Do This Before You Hunt Again

1. Go to a clear patch of your local ground (no power lines, no buried pipes or junk targets).
2. Turn your phone completely off.
3. Use your starting presets or use your last settings.
4. With your coil on the ground run a target over the top of the coil.
5. Listen, note exactly what the machine does in your real ground.
6. Write it down.
7. Repeat on every new claim or beach. The ground changes; your settings must change with it.

More importantly though is that as the temperatures change, so does your detector. In the desert as the sun comes up and warms the magnetic rocks on the ground, radiant energy is released. So always remember where you start in the morning with your setting is most likely not where you will be by the afternoon.

Do these tests and you'll stop guessing and start knowing.

These three questions stop being mysteries the moment you understand the ground is the boss and pure gold only lives in space.

Now you know what I know after five decades of swinging.

Let's go make the machine disappear into the dirt so the good stuff can talk.

BONUS QUESTION:

What is the best metal detector?

"The best metal detector in the world is the one that I have in my hands right now that I know everything there is to know about the detector, and I mean everything. Before you can trust you and your detector you need to know everything it is telling you, and that takes coil time."

Before we go any further, throughout the rest of this publication I will refer to testing your detector with different targets. This is not air testing; air testing doesn't work, it's not real. I've never found a gold nugget floating 4 feet off the ground. The closest you will be able to get to understanding target responses is to put your coil on the ground that you are going to be working in, set your sensitivity, ground balance, and threshold and swing a target above the coil. It's still not going to be a real signal (like a gold nugget that's been buried in the ground for millions of years) but it is a lot closer to the reality over air testing because you are hearing a target from a detector that is measuring the ground you are working right now.

I will also talk about your confidence piece. Most people will call this a test piece or test nugget. I don't. Targets that have been buried in the ground like gold nuggets for millions if not billions of years and relics that have been in the ground for over a century will not respond the same out of the ground as they did when they were buried.

I don't use test targets and I think that by the time you get to the end of this book you won't either. But I do carry a confidence piece. If I'm hunting in an area and my machine is too stable or I'm not hearing any tar-

gets I'll throw that piece out in front of me and just swing over it on the ground and it lets me know my detector is still working. At the end of this publication there's going to be a little thing about keeping your head in the game.

Chapter 2: Frequency — The Heartbeat That Finds Your Treasure

Frequency is the heartbeat of your detector. It's measured in kilohertz (kHz), but you don't need to memorize numbers to use it. All you need to remember is this:

Higher frequency = tighter, faster ripples → loves small targets close to the surface.

Lower frequency = bigger, slower ripples → punches deeper for larger, more conductive targets.

This is as techie as it's going to get. kHz is a unit of frequency measuring 1,000 cycles per second. The highest frequency detector right now is the XP-DEUS 2 with the HF 2 coil, which can prospect or treasure hunt up to 120 kHz. What that means is that in every second of time at 120 kHz the machine is cycling the frequency 120,000 times in that one second. That's a lot of energy. On the other side of that is an 8 kHz detector that cycles 8,000 times per second. On the other side of the spectrum is the Garrett ACE 250 which transmits at 6.5 kHz.

Extremely high energy like that 120 kHz expels itself pretty quickly. Think about it like this. There is a runner that is a sprinter and the fastest person on the face of this planet at the 100 yard dash. One day at a track meet the coach walks up and says, "Hey one of our long distance runners hurt their leg, I need you to step in." The runner we'll call him Mr. 120 says, "Sure anything for the team" and he goes to the starting blocks. Starting gun goes off he takes off like a flash and is way ahead of everybody else, until he runs out of energy around the 150 yard mark and gets passed by the rest of the 6.5 pack using less energy for the long haul.

Picture two fog horns on a foggy coast. One is high-pitched and screams right next to you — you hear it loud and clear, but it doesn't carry far. That's high frequency (think 40–71 kHz like the Gold Bug 2). It lights up tiny gold, small jewelry, fine chains, and micro nuggets right near the surface. Perfect for nugget patches, exposed bedrock dry washes, or shallow beach hunting where the good stuff is sitting in the top few inches.

Now picture the low, deep rumble of a big ship's horn that you can hear for miles. That's low frequency (5–15 kHz range). It travels deeper, grabs bigger coins, relics, large nuggets, and high-conductive targets buried way down. That's your relic hunter's best friend in old ghost towns or your deep-nugget machine in virgin ground.

I've hunted with both for decades. I've pulled sub-grain gold with a 71 kHz machine that a lower-frequency detector walked right over. I've also pulled a ½-ounce nugget at 24 inches with a lower-frequency setup that the high-frequency machine never even whispered at. Neither is "better." They're just different tools for different jobs.

Single-frequency vs multi-frequency — without the tech-speak

Single-frequency machines run on one heartbeat. You pick the machine (or sometimes the mode) that matches what you're hunting. Simple, reliable, and what most of us started with.

Multi-frequency machines (the newer ones that "run multiple frequencies at once, aka Simultaneous Multi Frequency or SMF for short") try to give you the best of both worlds in one swing. They're great — I use them too — but they still follow the same rules: the machine is blending those ripples and showing you what stands out. Don't let the marketing make you think it's magic. It's still ripples. It's still ground mineralization fighting back.

Over the last few years this has changed a great deal. SMF detectors can handle ground now better than in the past and work to eliminate off sig-

nals but they do have their limitations and that is why two of the top manufacturers Minelab and XP have the ability to choose a single frequency to drill it down over throwing everything into the ground.

Real-world examples from the field

- Nugget hunting in hot ground? High frequency + small coil = your best friend for pickers and fly specks.
- Relic hunting an old homestead or Civil War site? Lower frequency + larger coil = depth on buttons, bullets, and brass.
- Beach jewelry? High frequency in the wet sand for rings and chains that sank shallow. Be careful here, salt water can create a lot of interference.
- Park coin shooting? Middle-ground frequency so you don't miss the deep silver but still grab the modern stuff.

How to "bend physics" with coils (my favorite trick)

If you think about it what I've just written about above, I've said that you cannot use a high frequency machine and get any depth out of it and you're going to miss gold nuggets that are close to the surface with the low frequency machine.

Here's the part most people miss: you can change the rules a little by changing coils.

- Low-frequency machine acting too deep and missing tiny surface gold? Put a smaller coil on it. Suddenly it behaves more like a high-frequency machine for shallow small targets.
- High-frequency machine running out of depth? Put a larger coil on it. You'll pick up deeper targets without losing all the small-gold sensitivity.

I've done this for years on GPAA claims. Same machine, two coils in the truck, and I'm ready for whatever the ground throws at me.

Finding your detector's "sweet spot" fast

Every machine and every patch of ground has a sweet spot. Here's how I find it in ten minutes:

1. Use your starting presets or use your last settings.
2. Go to your test garden or a clear patch of the exact ground you'll hunt.
3. Ground balance, set your sensitivity and threshold and keep the coil on the ground.
4. Swing your confidence piece above your stationary coil and listen! You can make slight adjustments from there.

Do this once per new area and you'll stop guessing.

Bottom line from 55 years of swinging

The ground always wins the argument. Your job is to match the heartbeat (frequency) to the treasure you're chasing and the ground you're standing on.

Higher frequency for the small stuff near the top.

Lower frequency for the deep stuff that's been waiting.

SMF for averaging across the board and then a single frequency for drilling down on the targets.

Master this one concept and you'll instantly hunt smarter than 80% of the people out there.

Action Steps — Do This Before Your Next Hunt

1. Look up your machine's operating frequency (manual or quick Google — it's usually right on the spec sheet).
2. Grab two coils if you have them (small and large).
3. Go to your regular spot. Spend 15 minutes testing the same three targets in all available modes/frequencies.
4. Write down what worked best: "Small coil + high freq = screamed on the 0.1g picker at 3 inches."
5. Save that as a custom preset if your machine allows.
6. Next time you hit a new claim, repeat the test. It takes longer to set up camp than to dial this in.

Do this and frequency stops being a mystery and starts being your secret weapon.

The heartbeat is in your hand.

Now go make it sing.

Now you're asking, "What about Pulse Induction PI detectors, where do they fit in?"

I'll dive deeper into pulse induction in the next chapter. I bet you'll be surprised with this information. The two top PI detectors are extremely low kHz. The Minelab GPX 6000 comes in at 1.225kHz and the Garrett Axiom rolls in at 1.5kHz, respectfully. For PI detectors it is the delivery system onto the ground and how the signal is returned over VLF detectors that make PI machines stand out.

Chapter 3: The Ripple Effect — How Every Metal Detector Actually Works

Metal detecting is not complicated.

It may *feel* complicated when you pull that new machine out of the box and see all the knobs, buttons, and menu screens staring back at you. I've stood right there with folks on GPAA claims who look like they just opened a nuclear reactor manual. But here's the truth I've lived for fifty-five years: once you understand what's really happening under that coil, everything clicks. And it clicks the same way whether you're holding a $199 starter or a top-of-the-line flagship.

I want you to forget the complicated electromagnetic theory for a minute. I'm not going to throw kilohertz formulas or phase-shift graphs at you. Instead, I want you to picture this:

You're standing at the edge of a calm pond on a still morning. You pick up a small rock and drop it in.

What happens?

A ripple shoots out in a perfect circle. Now cut that ripple in half so you are only looking at the part going away from you. That ripple travels across the water until it hits something — a stick, a rock, the bank — and then the ripple bounces right back to you.

Now look at the returning ripple closely, it's smaller, less energy, not as big as it was going out.

That's it. That's exactly how your metal detector works in the field.

What we're doing when we turn on the machine is simply creating an electromagnetic field based on a frequency range. We send that signal into the ground, it creates those ripples, the ripples hit a target, the target absorbs some energy based on its conductivity, creates an eddy current and when the target absorbs all the energy it can, it sends a return signal back up to the coil. Transmit. Receive. Just like the rock in the pond.

Think of nothing more than ripples in a pond. It's just that simple.

Now let's break it down into the parts you actually need to know when you're out there swinging.

VLF The Transmit Side — Sending the Ripples

Inside your coil there's a transmit coil that's firing off that electromagnetic field. It's pushing energy into the ground at whatever frequency your machine is set to. Higher frequency means tighter, faster ripples — great for small targets close to the surface. Lower frequency means bigger, slower ripples that punch deeper for larger, more conductive targets.

But here's what matters in the dirt: as soon as that energy leaves the coil, it begins to decay. It gets weaker the farther it goes because of the ground and its minerals. That's why depth has limits on every machine — the ripples eventually fade out. But they may go further than you think, I'll get to that in a moment.

The Target — Where the Magic Happens

When those ripples hit something metallic, they create eddy currents inside the target. Think of it like the target soaking up some of that energy, getting "excited," and then releasing its own little ripple back toward the coil. The stronger and more conductive the target (silver, copper, big gold), the stronger the return signal.

Important point: that target has to be large enough to create a decent eddy current. A single microscopic piece of gold usually won't cut it by itself. But a nice little nugget or a coin? That's a whole different story.

VLF The Receive Side — Listening for the Return Ripple

Your receive coil — the second coil — picks up that returning signal. It tells the machine, "Hey, something's down there," and the machine turns that into a tone, sometimes a number on the screen, or both.

Simple. Repeatable. Works the same on every VLF detector ever made.

The Earth Is the Biggest Target You'll Ever Hunt

Here's the one thing most beginners miss, and I hammer this in every single class:

What is the largest target you will ever detect with your detector? The earth.

The ground itself is one massive target. And depending on where you are — hot red Arizona dirt, black sand beaches in California, mineralized clay in the Midwest — that ground is throwing back its own signals all day long. That's why your machine can sound noisy. That's why ground balancing matters so much.

Every time you swing, you're dealing with the earth's signal first, then whatever man-made or natural treasure is sitting in it. Keep that in mind and you'll stop fighting your machine and start working with it.

When in Doubt — Factory Preset and Think Ripples

I tell this to every new detectorist I meet: if you're confused, if the machine is screaming at you, if you just can't get a clean signal — hit the factory reset button and start over.

Wipe it clean. Set your threshold to that soft mosquito hum. Ground balance properly. Then go back to the ripple analogy. Drop the rock. Watch the ripples. Don't overthink the knobs.

Because here's what I've learned after five decades and thousands of hours in the field: the similarities between detectors are way bigger than the differences. Brand M, Brand F, Brand G, Brand W — they all create ripples, they all look for returning ripples, and they all have to deal with the earth being the biggest target.

Master the fundamentals and you can pick up any machine tomorrow and hunt with confidence.

Action Steps — Do This Right Now

1. Take your detector outside to a clear patch of ground (no power lines, no rebar plumbing, nothing but ground, target and coil).
2. Turn it on and hit factory preset/reset. Use your starting presets or use your last settings.
3. Set threshold to a faint, steady rhythmic hum you can just hear in your headphones.
4. Ground balance using whatever method your manual says (we'll cover this in detail in Chapter 6).
5. Swing low and slow with your coil on the ground while picturing the ripple every time the coil moves.
6. Bury a coin or a nail 4–6 inches deep and swing over it. Listen for the return ripple. Then bury the target even deeper. Keep burying it until you can barely hear anything. That's your baseline.

Do this drill ten times and the whole "complicated" feeling starts to disappear.

That's the ripple effect.

Once you get it in your head, you stop seeing a mysterious black box with a dozen buttons and start seeing a simple tool that sends ripples and listens for answers.

And that, my friend, is when the real fun begins — because now you're not just swinging a detector.

You're hunting.

Chapter 4: Sensitivity, Gain & All Those Other Buttons

We've got the ripples working. We've got frequency matched to the job. We've got threshold humming. We've got ground balanced so the earth shuts up.

Now we turn up the volume on what the machine can actually hear — without turning it into a screaming mess.

This is where a lot of folks go wrong. They see "Sensitivity" or "Gain" on the screen and think, "Crank it to max and I'll find everything!"

Wrong.

Sensitivity (or Gain — different brands call it different things, but it's the same job) is **receive** sensitivity. Not transmit power. The transmit side is fixed by the factory — that's the ripples going out. Sensitivity is how well the machine listens to the ripples coming back. Crank it too high and you're not getting more depth or more small gold. You're just letting the machine hear every little ground noise, EMI, hot rock, and phone in your pocket. It gets unstable, you get tired, and you miss the good signals buried in the chatter.

I've said it in every class and I'll say it again: **sensitivity is the most important setting on your detector** — but only when it's set to the ground you're standing on, not to some number the manual suggests.

The difference between sensitivity and gain (and why it doesn't really matter)

On some machines it's labeled Sensitivity. On others it's Gain. Doesn't matter. It's the receive control. Higher number = machine hears weaker signals. But those weaker signals might be the ground yelling, not gold.

Think of it like turning up the volume on a radio in a thunderstorm. Crank it too high and all you hear is static. Back it off to where the music is clear and steady — that's where you hear the song.

When to max it, when to back it off — field-tested rules

Here's exactly how I set sensitivity every single time, on every machine, in every patch of dirt:

1. Threshold set to that soft mosquito hum (or waver on digital).
2. Ground balanced perfectly (pump technique from Chapter 6).
3. Hold the coil about half an inch off the ground in a clean spot. "I carry a small piece of 1x4 wood with me to set the coil on and keep it stable. I call it my Sens Stick."
4. Start sensitivity/gain low (way down).
5. Slowly raise it until the threshold starts to get unstable — chatter, pops, false signals.
6. Back it off one or two clicks until it's rock-solid steady again.

That's your setting for this ground. Not 10/10. Not whatever the "expert" on YouTube runs. **Your ground.**

In mild beach sand I've run sensitivity fairly high if I can. In hot Arizona clay I might be at 4 or 5 on a 1–30 scale and still pulling 2–3 foot targets because the machine is stable and listening clearly. Crank it higher in that same dirt and it sounds like a fireworks show — and I guarantee you're walking over good signals. Even with that said, I always match the detector to the ground I am detecting.

Noise cancel, iron bias, recovery speed — demystified

These extra buttons are nice, but they're secondary. Get the big three right first.

- **Noise cancel**: If your machine has it, use it when you hear EMI (power lines, cell phones, other detectors). But the best noise cancel is prevention: turn your phone completely off or airplane mode and leave it in the truck. I've seen brand-new machines go crazy because someone's phone was in their back pocket.

- **Iron bias**: On gold and relic machines I usually run it low or off. It helps quiet iron, but it can also quiet small gold or weak signals that read close to iron. If you're coin shooting in a park, maybe bump it a little. If you're nugget hunting or relic hunting old sites — keep it low so nothing good gets filtered out.

- **Recovery speed**: Faster recovery helps in trashy areas — it resets quicker so you don't miss two targets close together. Slower recovery can give slightly better depth in clean ground. Most modern machines I run in the middle or slightly faster on GPAA claims with scattered trash. Test it with two targets buried close — you'll hear the difference fast.

"Set it and forget it" starting points for any machine

Once you've done the half-inch test in your actual dirt, save it as a custom preset if your machine allows. Then:

- Start every hunt: use your starting presets or saved preset → threshold → ground balance → sensitivity stable.
- Walk 20–30 feet. If the ground changes (hotter or milder), re-check sensitivity and ground balance.
- That's it. No chasing numbers. No "someone on the internet runs 28 sensitivity." Run what the dirt allows.

I've pulled more gold with "moderate" sensitivity in stable machines than I ever did cranking everything to max and fighting the chatter.

Action Steps — Do This Before Your Next Hunt

1. Go to your regular spot or a new claim.
2. Use your starting presets or last good settings → threshold mosquito hum → ground balance.
3. Hold coil ½ inch off clean ground.
4. Raise sensitivity/gain slowly until unstable, then back off one click for perfect stability.
5. Note the number and write it down for that ground type. Remembering that this will change throughout the day.
6. Swing over a buried test target at 6–8 inches. Listen clean and clear. Swing your confidence piece above your stationary coil, raising it until you have just a bump in the threshold. Listen to the signal and if needed, adjust from there.
7. Now crank sensitivity up 2–3 more clicks and swing the same spot — notice how noisy and tiring it gets.
8. Back it down to your stable setting. Feel the difference.
9. Do the same test for noise cancel, iron bias, and recovery speed if your machine has them. Find what keeps it stable and repeatable.

Do this drill and those "other buttons" stop being scary. They become tools instead of mysteries.

Sensitivity isn't about turning it to 11.

It's about turning it to exactly what the ground will allow — so every whisper from a real target comes through loud and clear.

Now your machine isn't just on.

It's listening.

And it's listening perfectly.

Chapter 5: Threshold — Your Ears Are the Most Important Knob

We've covered ripples and frequency. Now we get to the part that separates the people who find gold from the people who just swing a detector: your ears.

Threshold is the voice of your detector. Remember that you will hear things before you see them. Never discount a bump in the threshold even if you see nothing on the VDI.

It is not background noise you can turn off. It is not something you crank up until it screams. It is the constant, repeatable humming rhythmic signal that tells you everything that's happening under the coil — before the screen ever lights up, before the tone even breaks.

I've said this in every seminar I've ever given, and I'll say it again here: **if you turn your threshold off or set it to silent, you are walking over gold.** Period. End of story.

Most of the big, deep nuggets and a whole lot of the tiny ones I've found over fifty-five years were nothing more than a slight change in the threshold sound. Not a loud "ZIP-ZIP-ZIP." Just a whisper. A break in the rhythm. A tiny bump. If you're only listening for loud signals, you're missing a lot of the good stuff.

What threshold really is and why turning it off is usually a mistake

Threshold is the steady background sound your detector makes when it's running clean. On a VLF machine it should sound like a mosquito buzzing by your ear — just audible, steady, repeatable. It may be a little

fast, it may be a little slow, but it has rhythm. You lock into that rhythm like a drum beat.

On a pulse induction machine it's more like a wave — smooth, even audio in both ears.

When a target comes along, it breaks or changes that rhythm. Sometimes it makes the hum get louder, sometimes softer, sometimes it just hiccups for a split second. That change is your target talking. Turn the threshold off and you lose the conversation.

I've watched detectorists in my test garden walk right over a half-gram nugget because they had threshold at zero and were waiting for a big beep. Same nugget with proper threshold? It sang.

Setting it right — the "just audible hum" rule I teach in every class

Here's exactly how I set threshold every single time, on every machine:

1. Use your starting presets or use your last good settings.
2. Ground balance properly (we'll get there in Chapter 6).
3. Turn threshold up until you can just barely hear it — that soft mosquito hum in your ears. If you have one of the newer tech detectors that is all digital, you will hear what is more like a straight line hum. Back it off to get a little waver. You will never get the mosquito with the digital.
4. If it's too loud and annoying, back it down one click. If you can't hear it at all, bring it up one click.
5. Swing the coil. The hum should stay perfectly steady until something real interrupts it.

That's it. That's the sweet spot I depend on. I call it finding the "wavelength" I'm so dependent upon.

Do this and even the faintest changes become signals that came out of the ground as good targets.

How threshold tells you about ground changes before the screen does

Swinging with a proper threshold is like having a second set of eyes on or in the ground. The threshold will start to break or get choppy before the screen shows mineralization change. It will get quieter or louder when you hit hot rocks or black sand. It will tell you when your coil is too high or when you're swinging too fast or too slow and if you're wavering from having it on the ground where it should be to lifting as you swing. No matter what—if you have a repeatable signal, dig it unless you know 100% that it is junk.

In extreme ground I've used the threshold to hear variations in mineralization and dial them out before the machine even complains on the screen. That's real-time information you can't get any other way.

"Can you hear me now?" — Audio mastery from my magazine and podcast tips

When everything is right — coil on the ground, ground balanced, sensitivity set to the actual dirt you're standing on, and threshold at that perfect mosquito hum — your detector will start talking to you clearly.

It will literally ask, "Can you hear me now?"

And when it does, the answer is always, "You bet I do."

I've walked over nuggets myself because I had the threshold wrong. I've had students do the same until I made them set it exactly like this. Once they hear that rhythm and start catching the whispers, their finds go up dramatically.

Headphones are non-negotiable for most of us. I've run Nugget Busters and Black Widows for years because they let me hear those faint changes. If you wear hearing aids, we'll cover solutions in the advanced section, but the principle stays the same: you have to hear the voice.

Action Steps — Do This Right Now (Before Your Next Hunt)

1. Take your detector to a clear patch of ground (your backyard test garden is perfect).
2. Use your starting presets or last good settings → ground balance → adjust sensitivity → set threshold to that just-audible mosquito hum (or waver on digital).
3. Coil on the ground, swing slow. Listen for the steady rhythm.
4. Swing your confidence piece above your stationary coil, raising it until you have just a bump in the threshold. Listen to the signal and if needed, adjust from there. Note exactly how the threshold changes.
5. Turn threshold all the way off and swing the same spot again. Notice how much you miss.
6. Put it back to the mosquito setting. Swing again. Feel the difference.
7. Write down the exact threshold number that worked in that ground. Do this for every new area — the ground changes, your settings must change with it. Remember this will change throughout the day!

Do these steps and your ears become the most powerful knob on the machine.

Threshold is not just another setting.

It is the voice of your detector.

Learn to listen to it and it will never lie to you.

Now you're not just swinging.

You're having a conversation with the ground.

And the ground has been waiting a long time to tell you where the good stuff is.

Chapter 6: Ground Balancing — Making Peace with Hot Ground

Now it's time to make the biggest target in the entire field — the earth itself — shut up and disappear.

That's all ground balancing is.

Nothing more. Nothing less.

Every patch of dirt is different. Some is mild like a beach in Florida. Some is so hot it'll make your machine sound like it's having a meltdown — red Arizona clay, black sand in California, magnetite-laden washes in Nevada, salty beaches, iron-rich Midwest soil. That ground is throwing its own signal back at your coil every single swing. If you don't balance to it, your detector is fighting the earth instead of hunting treasure.

I've stood on GPAA claims where the ground was so mineralized that an unbalanced machine would false on nothing but air. I've also watched students who learned to balance properly pull nuggets the guy next to them swore weren't there. The difference is always ground balance.

Manual vs Automatic vs Tracking — when to use each (real GPAA claim examples)

Every machine gives you options. Here's how I use them in the real world:

Manual ground balance — My go-to on hot, consistent ground.

You control it completely. On machines with a dial or buttons, you pump the coil and tweak positive or negative until the threshold is perfectly steady. I love this on the Gold Bug 2 or older machines because I can frac-

tionally fine-tune to any ground. On a nasty Arizona claim that's been pounded for years, and because I am always keeping the detector manually ground balanced by making small tweaks all the time, I hear every whisper. A manual ground balance detector is not for everyone and I understand that. You have to commit to keeping the balance finely tuned with every swing of the detector. And you have to hear and respond to the threshold every time there is a change that is not a target sound. It's the ground talking to you.

Automatic (or "auto") ground balance with ground grab

This is a ground balance the more people than I would like get confused. And if you have not bought your detector yet, make sure of what you are buying. The Automatic Ground Balance does not mean a constant ground balance on many detectors.

What it really means is that when you press the button the detector goes from a MANUAL ground balance to auto for re-balancing. When you let go of the button, it goes back to being a manual ground balance.

Process: Pump the coil a few times and the machine averages the ground under it. When I release the button, I check the ground with a couple of pumps of the coil and if there is a slight variance in the threshold, I'll use the ground grab to bring everything to the center line and go detect. After that I will use the ground grab over re-balancing the machine from scratch. Works awesome on the Nokta AU Gold Finder.

Auto Ground Tracking

When the ground is changing fast.

The machine constantly adjusts GB as you walk. Perfect for patchy ground, and any area where the ground is unstable that go from mild to hot constantly, or after a rain when mineralization shifts. On the Garrett 24K I tell folks it's almost "turn on and go" — Start out in tracking and

keep an eye (and ear) on it. Just don't trust it 100% in extreme hot spots; it can average out small targets if you're not careful. And when you start hearing changes, do a quick ground grab and keep detecting.

Here's my rule on GPAA claims:

Start with auto or tracking to get going. Switch to manual (if your detector has a manual GB) or ground grab when you hit a hot patch and want every tiny edge. Test it. The ground will tell you which one wins.

The "pump the coil" technique done right

This is the one thing I make every student do in the field until they can do it in their sleep:

1. Hold the coil 3–4 inches above the ground in a clear spot (no targets).
2. Pump the coil straight up and down — no higher than 4 inches each way — nice and smooth.
3. Listen to your threshold. It should stay perfectly steady with no rise or fall.
4. If it rises or falls, adjust the ground balance knob or buttons while pumping the coil up and down until the threshold is smooth and equal going up and down.
5. When it's rock-solid steady, you're balanced. The earth under your coil just disappeared.

Your ground balance will change and you have to stay on top of it, even with an auto-tracking detector. The moment your threshold becomes more than the setting you dialed in, check and adjust your ground balance. Sometimes you may only do this once in a while, and in other areas I've had to adjust after every couple swings of the coil.

Common mistakes that make your machine scream for no reason

- Pumping too fast or too high — the machine can't read the ground properly.
- Not checking at the end of every swing in manual — it drifts and you start chasing ghosts.
- Forgetting to re-balance after you change sensitivity or frequency, and most importantly, after every target.
- Balancing over a target by accident (yes, it happens — get away from the target).

I've done every one of these myself over the years. Learn from my mistakes.

The payoff — when the ground goes silent

When you get ground balance right, the machine goes quiet. The constant chatter stops. The threshold hum stays smooth. Now every break in that hum in the threshold is a real target — not the dirt yelling at you. That is unless it is a hot rock or cold stone.

That's when you start finding the stuff others miss. That's when the detector disappears into your hand and you're just hunting. Detecting.

I still get a kick out of watching a new detectorist nail their first perfect ground balance on a claim. Their eyes light up because suddenly the machine makes sense.

Action Steps — Do This on Your Next Outing

1. Go to your regular spot or a new GPAA claim.
2. Use your starting presets or last good settings → set threshold to mosquito hum.
3. Find a clear patch of ground.
4. Practice the pump technique in manual until the threshold is dead steady.

5. Switch to auto — pump and compare.
6. Switch to tracking — walk 20 feet and see how it handles changes.
7. Swing your confidence piece above your stationary coil, raising it until you have just a bump in the threshold. Listen to the signal and if needed, adjust from there.
8. Repeat every time the ground feels different. Make it a habit like checking your shoelaces.

Do this and hot ground stops being your enemy and becomes just another part of the game.

You've now made peace with the earth.

The ripples are clean.

The threshold is talking.

The ground is quiet.

Everything else is treasure waiting to be found.

Chapter 7: Coil Control — The #1 Skill That Separates Finders from Walkers

We've got the ripples, the frequency, the threshold humming, the ground balanced, and the sensitivity set to what the dirt will actually allow.

Now it's time for the one thing that turns good detectorists into great ones — and the one thing I see more people mess up than anything else.

Coil control.

It's not flashy. It's not a new button or a fancy mode. But after fifty-five years of swinging, I can tell you flat-out: coil control is the #1 skill that separates the people who find gold (and relics and rings) from the people who just walk around waving a detector.

Keeping your coil on the ground is imperative to success.

And I mean *on the ground*. Sliding it. Wearing out the coil cover. If you have a hard time picturing this, go watch the short video I made showing two of the best-selling VLF detectors completely missing a half-gram gold nugget with the coil less than two inches off the ground. Same nugget, same spot, same settings — lift the coil even a little and it goes silent. Put it back on the ground and it screams.

Every inch the coil is above the ground is an inch of depth you just threw away.

I've said it in every class and I'll say it again here: **Coil Covers are cheap. Missing gold nuggets isn't.**

When you're running the coil right, you're going to generate a lot of noise from rocks, sticks, and dirt. That's why headphones are non-negotiable for me. You need to hear everything.

The "low and slow" gospel — proper swing speed, overlap, and height

Low and slow. That's my mantra. I've been saying it for decades.

- **Height**: Coil flat to the ground, parallel at all times. No bouncing, no tipping on the edges. Slide it like you're mowing the lawn with a razor blade.
- **Speed**: Slow. So slow that if you're in a hurry you'll get frustrated. If you swing too fast the machine will literally tell you — the threshold breaks on the return swing or you overrun the target and never hear it clean. I tell students, "If you're going too fast it'll tell you... slow down, that's your detector screaming at you."
- **Speed**: Fast. On the other side of that if you're going too slow you're going to hear it in your threshold as well. When you start to hear chatter speed up a little bit and if the chatter goes away, or I should say as soon as the chatter goes away that's the ground telling you your coil speed is right.
- **Overlap**: At least 50% — half the coil width on every pass. No gaps. Think of it like painting a wall — you overlap each stroke so you don't miss a spot.

Do this and you stop missing targets that are literally right under your coil.

Figure-8

Figure-8 coil swings have for some reason become popular these days. I don't know how this became a thing, and I do not recommend the 8 swing, too many chances to miss, well, everything. I'll use an 8 to hear the ground in some areas before I ever start setting up my detector. If I'm on new ground that I've never heard before and I do a Figure 8 I'll be crossing my own signals as I go through the 8. It really lets me hear the energy that I'm putting into the ground and gives me a good idea of where I may end up with my sensitivity and ground balance before I even start. Let's just call that a pro tip and leave it at that.

Most of the time I run a straight-line "lawnmower" pattern — back and forth, overlapping, keeping the coil dead level. Perfect for systematic grid work (we'll cover grids in Chapter 8) and covering big patches without missing anything.

My rule: Side to side for complete coverage then 90° for tight spots.

Why people miss targets right under their coil

I've watched it a thousand times on GPAA claims:

- Coil floating 2–4 inches up (especially when they get tired)
- Swinging too fast and outrunning the signal
- Swinging too slow and not recognizing the ground sounds and targets
- Tipping the coil so only half the field is working
- No overlap — leaving "railroad tracks" of missed ground between swings
- Looking at the screen instead of listening and feeling the rhythm. If you are in one of my classes and I catch you spending all your time looking at the VDI, I'll probably put a bag over your display.
- At the end of the swing lifting their coil like they're using a scythe for cutting grass. Coil on the ground from end to end of

your swing. Quit trying to detect airplanes.

All of those are cured by one thing: low and slow with the coil on the dirt.

My favorite drill from the Metal Detecting 101 field sessions

This is the one I make every student do until it's muscle memory:

1. Go to your backyard or a clear patch on a claim (no power lines).
2. Bury three targets at different depths — a small nugget or lead piece at 3 inches, a coin at 6 inches, a larger piece at 6–8 inches.
3. Set your machine perfectly (threshold, ground balance, sensitivity).
4. Practice your swing: low, slow, 50% overlap, coil flat and sliding.
5. Close your eyes or look away from the screen and just listen to the threshold rhythm.
6. When you hear a break, stop, back up, come at it from different directions. What I call the East to west and north to south to the 90° rule.
7. Dig it only after you've confirmed a repeatable signal from multiple angles.

Do ten targets like this and you'll feel the difference immediately. Your swing becomes smooth, repeatable, and deadly accurate.

I still do this drill myself when I get a new machine or hit a new patch. It keeps me honest.

Action Steps — Do This Before Your Next Hunt

1. Take your detector outside right now.

2. Put the coil on the ground and slide it back and forth — feel it wearing on the cover. That's where it belongs.
3. Practice 50 slow, overlapping straight-line passes across your yard. Keep the coil dead level.
4. Bury 3 test targets and run the full drill above. Time yourself — it should take longer than you think because you're going slow.
5. On your next real hunt, consciously check every 3 minutes: "Is my coil on the ground? Am I overlapping? Am I slow enough?"
6. Wear out a coil cover this season. Celebrate it — it means you're doing it right.

Master coil control and everything else gets easier. The machine disappears. The ground opens up. And the targets you used to walk over start jumping into your pocket.

This isn't the sexy part of metal detecting.

But it's the part that actually finds the gold. And finding gold is sexy!

Low and slow, coil on the dirt, overlap every pass.

Do that and you stop walking.

You start finding.

Chapter 8: Grid Patterns & Systematic Hunting — Stop Random Walking

We've got the machine tuned. Coil on the ground. Threshold talking. Now it's time to stop walking around like a lost tourist and start hunting like someone who actually expects to find something.

Random walking is the biggest mistake I see out there. Guy swings for an hour, covers a football field, finds nothing, and says "this spot is hunted out."

No. That spot probably still has gold — you just didn't hunt it right.

Why gridding beats "hunting where you feel like it" every single time

When you find that first good signal — whether it's a nugget in the desert, a relic in a ghost town, or a ring on the beach — stop. Right there.

That one piece tells you something important: **there's more here.** Gold and treasure don't usually travel alone. They hang out in patterns. So instead of wandering off looking for the next thrill, you establish a grid right on top of that find and hunt it systematically.

I've done this on GPAA claims more times than I can count. I hit one nugget, grid the area, and walk away with a dozen more that everyone else walked right over. Same claim, same day, same ground — the difference was the grid.

Box grid, lawnmower, spiral — which to use where

Here's how I actually do it in the dirt:

Lawnmower pattern (straight-line north-south) — My everyday workhorse.

Mark your starting line (more on marking in a minute). Swing back and forth like you're mowing the lawn — 50% overlap, coil on the ground, low and slow. Cover the whole box north to south. When you finish, turn around and do the exact same area east to west. Why both directions? Because the gold could be lying in a different orientation. One direction the eddy currents hit perfect. The other direction it might be silent. I've seen it dozens of times.

Box grid — For bigger patches or when you've got a hot zone.

Stake or mark a square or rectangle (10x10 feet, 20x20, whatever fits the terrain). Hunt the entire box north-south, then east-west. Then diagonally I'll call this SW to NE if the ground is really patchy. Perfect for nugget patches in the desert or relic fields.

Spiral pattern — When you get a signal and want to tighten up fast.

Start wide around the signal and spiral in tighter and tighter until you pinpoint. Also great on beaches or open parks where you want to work outward from a hot spot without missing the edges. I use this a lot when I first hit a new area and want to feel out the concentration.

My rule on GPAA claims: Start with the lawnmower or box when you find that first piece. Switch to spiral to zero in. Then expand the whole grid outward if you keep finding more.

Marking your grid without looking like a surveyor

Nobody wants to look like they're laying out a construction site. Here's what we actually do in the real world:

- **The chain trick** (NOT my favorite but what a lot of old-timers

still use): Hook a short length of light chain to the back of your pants or belt. As you walk and turn at the end of each pass, drop or throw the chain behind you. It marks your line perfectly and drags with you so you always know exactly where you've been and where you're going next. No flags blowing over, no stakes to trip on.

- **Poker chips, rocks, sticks, or hats**: Drop something visible at the corners and every few passes. Move them as you expand.
- **Your own footprints**: In soft ground or sand, just follow your tracks — they show you exactly what's been covered. Pro tip: every few feet drag your inside foot across the ground a bit, that way when you make your turn, you can see exactly where your last line ended.

Keep it simple. The goal is to know you've hunted every inch, not to win a landscaping contest.

How I've found nuggets others walked right over on the same claims

I'll never forget a day on a popular Arizona GPAA claim. Three guys had been there all morning, swinging fast, covering ground, talking about how it was "worked out." I pulled in, found one small picker, established my little 10x10 grid, hunted it north-south, then east-west. Pulled seven more in the next hour — all in the same area they had just walked. Same machines, same ground, same day. The difference? They were random walking. I was gridding.

Another time in a dry wash with exposed bedrock I hit one nice piece, gridded it, and expanded three times because every time I found another I moved the box over. Ended up with a nice little pocket that had been sitting there for who knows how long.

That's the power of the grid. It forces you to slow down, put your head in the game, and believe there's more where that first one came from.

Action Steps — Do This on Your Next Outing

1. Next time you hit a signal and dig a target, stop and mark a small grid around it (start 10x10 feet).
2. Hunt it north-south with full overlap, coil on the ground.
3. Turn and hunt the same grid east-west.
4. If you find more, expand the grid and repeat.

When you find more gold stop for a minute and look to see if there is a line that the gold was flowing during a distribution stage of massive water. That is a TREND and they must be followed.

1. Practice the chain trick (or poker chips, rocks/footprints) in your backyard first so it feels natural.
2. On a full hunt, pick one small area and grid it completely before moving on. Compare how it feels versus your old random style.
3. At the end of the day, look at what you covered. You'll be amazed how much more thorough it is.

Slow down. Put your head in the game. Believe in the grid.

Random walking is for tourists.

Gridding is for finders.

Do this and the spots everyone says are "hunted out" will start talking to you again.

Chapter 9: Tones, Discrimination & The Dig-or-Don't Decision

We've got the machine tuned, coil on the dirt, grid laid out. Now the detector starts talking to you — and you'd better learn the language fast.

I've been saying this for decades and I'll say it again here: **I'm not a big fan of discrimination.** I'm an all-metal guy who believes in digging every target. Completely removing hot rocks using ground balancing and using discrimination for some iron targets all the time can cause you to miss good targets that are masked by junk or carrying enough minerals to sound bad when they're actually valuable.

All true gold prospecting detectors are ALL METAL period. Because there is no such thing as pure gold in nature, gold nuggets can have both ferrous and non-ferrous metals as a rider.

There is no truly accurate nonferrous discrimination on gold detectors. At best it works "kinda." Trusting it is dangerous and can lead to missed gold. Now there are slight exceptions to that rule and that is in an Iron ID that when you press the button, it will groan when passed over junk iron. BUT you cannot trust this. I have a gold nugget that the gold is shot through magnetite. Yep, a ferrous gold nugget that every detector that I have tested shows it as junk.

And if you didn't catch it above, **HOT ROCKS (a part of the ground) are Ground Balance and Junk (The stuff humans left behind) is Discrimination. PERIOD.**

Learning your machine's language (high tone, low tone, broken signals) — COIN AND RELIC DETECTING ONLY!

Every crossover detector has its own dialect. Some use high tones for non-ferrous, low tones for iron. Some use varying pitches, volume, or broken/choppy sounds. Your job is to learn exactly what your machine is saying in the ground you hunt.

- High, clean, solid tone → usually good stuff (coin, ring, nugget).
- Low, grunt, iron tone → usually trash, but not always (big rusty iron can give false high tones).
- Broken, choppy, or warbling signal → could be a good target on edge, masked by iron, or a hot rock. This is where experience pays off.

I tell every student the same thing: the only way to learn the language is to dig. A lot. Keep a little notebook or voice memo on your phone. "6-inch broken high tone in hot ground = 0.3g nugget." Do that 200 times and the machine stops being a mystery and starts being a partner.

When to discriminate... and when to turn it off completely

Here's my real-world rule:

- In a trashy park or beach full of bottle caps and pull tabs? You can use a little discrimination to quiet the chatter so you don't go crazy. Just don't trust it 100%.
- Nugget hunting, relic hunting old sites, or anywhere the good stuff might read close to iron? Turn discrimination completely off. Run all-metal. Let it talk.

On the AU Gold Finder I tested both DISC 1 (Fast) and DISC 2 (Deep). They have their place, but I never became dependent on them. Same on every machine I've ever run. Discrimination is a tool, not a crutch.

If the ground is hot and mineralized, discrimination gets even less reliable. That's when I go all-metal on my crossover detectors and let my ears do the work.

There are two crossover detectors that I do trust because of the NOTCH Discrimination system. I have been using the XP-DEUS II and the HF2 coil recently and I have found that in the Gold Field program I can use notch to remove iron junk and I have tested on a few non-ferrous junk targets but that still scares me a bit with non-ferrous targets. In the cases of ferrous junk targets, like a nail, I'll notch out just a small part of the nail not the complete target. And on the gold side, the Minelab Gold Monster 2000 has a fairly good iron disc baked in when using Echo Wave.

I am positive that there are other companies working on making discrimination better and time will tell.

"If in doubt, dig it out" — my personal philosophy with real stories

This is the one sentence I've repeated more than any other in fifty-five years.

If the tone is questionable, if it's a one-way hit, if it's mixed high and low, if it doesn't repeat perfectly from every angle — dig it.

I've dug thousands of nails, bottle caps, and hot rocks because of that rule. I've also dug a lot of nice gold and relics that would have been left in the ground if I'd trusted the screen or the discrimination setting.

One day on a GPAA claim a guy next to me walked off a signal because "it read ferrous." I dug it. Nice little picker. Same ground, same day, same machines. He trusted the disc. I trusted the dig.

Another time in a relic field I got a choppy broken tone that most people would have ignored. Dug it — turned out to be a beautiful old silver but-

ton with the iron spike still in it. The discrimination would have called it junk all day long.

Trash to treasure mindset

Every piece of trash you dig teaches you something. Every nail, every can, every piece of wire is a lesson in tones and how the ground affects signals. Pretty soon you start seeing the pattern: that particular broken tone in this particular ground usually means a small nugget on edge. That warble over there often means a deep coin.

The trash becomes your teacher. The more you dig, the better you read the language. The better you read the language, the more treasure ends up in your pocket.

That's the trash-to-treasure mindset.

Action Steps — Do This on Your Next Hunt

1. Run your machine in all-metal (discrimination off) for the entire session.
2. Every signal you get, note if it's clean or broken, high or low, repeats from all directions.
3. Dig it. All of it. Even the obvious iron.
4. After you recover the target, write or record: depth, tone description, what it actually was.
5. Do this for 50 targets.
6. In trashy areas only, test a light discrimination setting on a few signals — then dig them anyway and compare. See for yourself how often it lies.
7. Next hunt, try the same spot with discrimination on vs off. Compare your finds.

Then give a long and hard consideration on whether or not you are willing to risk missing gold. I've done a lot of testing over the years and I keep going back to digging every repeatable signal.

Dig it all. Learn the language. Trust your ears and your experience more than any discrimination circuit ever made.

Because the gold that's been missed for years wasn't hidden by the dirt.

It was hidden by the "don't dig that" decision.

If in doubt... dig it out.

You'll be amazed what starts showing up.

Yet when I boil it all down, I use gold detectors for gold and CRT detectors to Coin-Relic & Treasure hunting with the exception of the XP-DEUS II and only with the HF2 coil. A true crossover detector.

Chapter 10: Digging Tools, Recovery & Leaving No Trace

We've tuned the machine, gridded the patch, learned the tones, and decided to dig.

Now the fun part — actually getting that target out of the ground without tearing up the landscape or spending an hour on one hole.

I've been digging targets for fifty-five years. I've dug in desert hardpan, beach sand, muddy riverbanks, grassy parks, and everything in between. The right tool and the right technique make all the difference between a quick, clean recovery and a torn-up mess that gives every detectorist a bad name.

Best plugs, tools, and techniques for different soil types

I don't have a one-and-done pick. I pick my pick according to the ground I'm detecting.

- **Hard, dry, mineralized desert or clay** — Heavy-duty pick with a wide blade. I've used everything from a custom one I made out of the tip of a number 2 shovel and a plasma-cut leaf spring welded to a wooden handle (still going strong after a decade and thousands of targets) to an Apex Badger or Mad Mining Stealth. The key: heavy duty and wide enough to pull material out of the hole with ease. Least energy, maximum dirt moved.
- **Soft sand or loose gravel** — A good sturdy scoop does most of the work. I generally use Doc's new scoop (the one with the riffle for everything) or the GPAA scoop.

- **Turf or grass** — Slot recovery all the way. Cut a narrow slit next to the target, pry it open like a book, pull the material onto a drop cloth, recover the target, replace the dirt, and step it closed. Roots stay intact, grass doesn't die, and nobody can tell you were ever there.

Always carry a rare-earth magnet. It's a must for quickly pulling junk iron out of your dig pile so you're not chasing false signals.

How to dig fast without destroying the ground (GPAA ethics)

Two scoops always.

With today's sensitive machines you're going to get signals on your hands, your scoop, your knee — everything. Get the target out of the hole and into one scoop, then halve the material back and forth between two scoops until the target is in your hand. Fast, clean, no guessing.

The GPAA way is simple: dig only what you need, replace every bit of dirt, and leave the spot looking untouched. That's not just good manners — it's how we keep claims open for everybody.

And as always... **FILL IN YOUR HOLES.** Every single one. No exceptions. I don't care if it's 100 degrees and you're tired. Fill it, tamp it, make it disappear. That one habit separates responsible detectorists from the ones who get land closed to all of us.

Reclaiming your dig holes on a claim is not a courtesy, it is a Federal Law.

Pinpointing like a surgeon

Once you've got a solid tone, pinpoint first — don't just start digging a crater.

Use your machine's pinpoint mode or the edge of the coil. Wiggle, listen for the strongest signal, mark the spot with your boot or a rock. Then

dig beside it, not on top of it. The two-scoop method I mentioned earlier is pinpointing in action — you're isolating the target without destroying half the grid.

If you run a separate pinpointer (highly recommended once you're serious), use it after the initial plug or slot. It'll save you minutes per target and keep your holes tiny.

Cleaning, photographing, and logging your finds

Take the time to do it right — it turns a good day into a great story and great data.

Please note that I DO NOT clean my gold nuggets except to brush off some of the dirt or clay. That's it. With coins and relics I follow what I have below.

- Clean gently — water and a soft brush for coins and relics, dry brush for nuggets.
- Photograph in place if possible (with a coin or ruler for scale), then again cleaned.
- Log it: date, GPS or claim name, depth, detector settings, what it was. I've gone back to logged spots years later and found more because I knew or learned the trend. The pattern.

That logbook or phone note becomes your personal treasure map over time.

Action Steps — Do This Before Your Next Hunt

1. Inventory your tools: pick/scoop matched to your main hunting grounds, rare-earth magnet, drop cloth for turf, pinpointer if you have one.
2. Practice the two-scoop recovery and slot method in your backyard on test targets. Time yourself — get fast and clean.

3. On your next hunt, consciously say out loud before you walk away from every hole: "Filled and tamped." Make it automatic.
4. Photograph and log your first five finds of the day — no exceptions. You'll thank yourself later.
5. At the end of the day, walk your grid one more time and double-check every hole is invisible.

Dig fast, dig clean, dig responsible.

The tools are simple. The ethics are non-negotiable.

When you leave a spot better than you found it, you're not just recovering treasure — you're protecting the future of this hobby for every one of us.

Now get out there, dig smart, fill every hole... and keep the claims open for the next guy.

Chapter 11: Where to Hunt — Research, Claims & Hot Spots That Actually Produce

We've tuned the machine, mastered the grid, learned to dig clean, and now the big question every new detectorist asks me on a claim or at a seminar:

"Where do I actually go?"

The answer is simpler than most people make it. There's gold, relics, rings, and coins still out there — you just have to know how to get legal access and how to research so you're not swinging where ten thousand other people already did.

Public land, private permission, GPAA/LDMA claims — the smart way

Public land (BLM, National Forest) is open to casual detecting in most places, but check the rules — some areas are off-limits near active mining claims or archaeological sites. And both the BLM and USFS can seasonally close an area. Sometimes I have to remind them that they need to reopen an area. Always verify with the local BLM office or app before you drive out. It's free, but it's also the most hunted ground on the planet.

Private permission is gold if you do it right. Knock on the door, be polite, offer to split finds and just show respect. I've gained access to some incredible old homesteads this way. One simple rule: leave it better than you found it and you'll be welcome back.

The GPAA has the Code of Ethics in the GPAA printed Mining Guide. I copied the COE with my scanner and carry copies with me all the time.

I'll knock on a door, introduce myself with my business card and the COE in my hand. That has gotten me onto more ground than anything. The card is my identity. My code of ethics is my pledge to them.

Then there's the smartest play for serious hunters: **GPAA and LDMA claims**. As a member you get exclusive access to hundreds of claims and leases across the country — East Coast, West Coast, desert, mountains. No claim-jumping, no competition from the general public. That's why we built the Online Mining Guide.

Here's exactly how I use it (and how I teach every Gold Trails group):

1. Log in at the GPAA member portal.
2. Search by ZIP code, city/state, or claim name.
3. Read every member report on that claim — dates, what was found, where, what worked. Those reports are pure gold intel from people who were just there.

East vs West claims are different animals. Eastern ones are often smaller, woodsy, relic-heavy. Western ones are bigger desert or mountain claims perfect for nugget patches. Pick your poison and stick to it for a while.

Reading old maps, newspapers, and history for lost treasure

This is where the real edge comes from. The old-timers — the sourdoughs — left clues everywhere.

- Grab historical topographic maps (old USGS quads) and overlay them on Google Earth. Look for old roads, mine symbols, dry washes that used to run year-round.
- Newspapers from the 1800s–1950s (online archives like Newspapers.com or local libraries) are loaded with "lost mine" stories, stagecoach robberies, and "gold found at XYZ ranch."
- County records, old claim filings, even ghost town histories —

they tell you where people actually lived and worked.

I still do this every time before a new Gold Trails trip. Three weeks of research can put me on a patch nobody's touched in decades.

Beach, park, ghost town, nugget patch — terrain-specific tactics

Every ground type plays different. Here's how I hunt them:

- **Nugget patches & washes**: Study the contours. Bedrock traps gold. Look for cracks, look for everything, and do not fall for the myth that all the gold is behind boulders and inside bends where water slowed. On GPAA claims like THREE R 2 in California I tell folks to walk the high banks and grid the gravel bars. After rain is prime time.
- **Ghost towns & relic sites**: Old foundations, privy pits, wagon roads. Slow grid, all-metal, dig everything questionable. The "trash" is often buttons, coins, or tools worth more than gold.
- **Beaches**: Wet sand after storms or low tide for jewelry. Dry sand for coins and rings lost in towels. Work the towel line where people sit.
- **Parks & schoolyards**: Playgrounds, under old trees, picnic areas. Early morning before the crowds. Discrimination if it's trashy, but still dig the iffy ones.

The key on every terrain: research first, then grid like your life depends on it.

My favorite "sourdough" spots and why they still pay

The old-timers called experienced prospectors "sourdough." I've got a few spots I return to on Gold Trails and GPAA claims that the sourdoughs knew about — and they still produce because most modern hunters random-walk and quit too soon.

Rye Patch, Nevada — famous for nuggets, but the real patches are the ones you find by re-gridding the "hunted out" zones after reading old reports.

Arizona desert washes — the ones with the right geology (Exposed Bedrock) and water history.

California claim spots where we study the contours together and suddenly everyone starts finding.

Here is a huge pro tip; Unless you can see bedrock, stay out of washes! Every bit of lightweight junk finds its way to the washes and yes so does some small gold. Plus you can have feet of overburden that your detector will never get through to the gold that has settled in and sank through the sands.

The bigger nuggets for the last 1.8 billion years or so that are on the higher fan flats or resting nicely on a bench deposit have been there since it was stranded there.

They pay because the research puts you in the right 10x10 foot box that everyone else walked past.

Action Steps — Do This Before Your Next Hunt

1. If you're not a GPAA member yet, join — then log into the Online Mining Guide today and pick one claim within driving distance. Read every report.
2. Open Google Earth. Find your nearest public land or GPAA claim and overlay an old topo map. Mark three potential hot zones.
3. Spend one evening on newspaper archives searching your county + "gold found" or "lost mine" from 1880–1950. Save the best leads.
4. For your next outing, choose ONE spot and commit to

gridding it completely before you leave.

5. If going private land, practice your permission script: "Hi, I'm a responsible detectorist with the GPAA. Would you mind if I hunt your back field? I'll fill every hole and show you anything interesting I find."

Do the research. Stick to the spot. Use the claims we maintain for you.

The treasure isn't gone.

It's just waiting for someone who knows how to look in the right places the right way.

That someone can be you — starting this weekend.

Chapter 12: Advanced Field Tips & My Best War Stories

We've covered the fundamentals. Now let's get into the stuff that only comes after you've put in the swings — the advanced tricks, the curveballs the ground throws at you, and the war stories that still make me grin like a kid with his first detector.

These are the things I've learned the hard way over fifty-five years so you don't have to.

Hot rocks, black sand, and other curveballs I've beaten

Hot rocks are one of the most common questions I get. A hot rock is simply a rock that has higher magnetic or metallic properties than the surrounding ground. Your machine sees it as a target and can drive you crazy if you don't know how to handle it.

Hot mineralization is different — it's the overall ground throwing signals. Learn to tell them apart and you're halfway there. I can easily make any detector work in hot ground; that's just dialing it in.

Black sand is another beast. I've spent days on claims tracking black-sand paystreaks with a Garrett Gold Master 24K because that machine's black-sand tracking mode lets you follow the iron mineralization right to where the gold likes to hide. Same on the Gold Bug 2 — higher frequency annual ground balance and a solid hot rock identifier (signal is one sided and not repeatable) quiets the hot rocks so the faint gold whispers come through.

Cold stones, iron soil, magnetite — I've beaten all of them the same way: test garden practice (more on that in a minute), proper ground balanc-

ing, and patience. The ground will throw curveballs. Your job is to stay calm, re-balance, and keep the coil on the dirt.

However in this list there is one item that you are going to have to get used to hitting and you do not ever want to completely ground balance them out and that is a cold stone. Cold stones are non-ferrous and are denser than the surrounding ground which makes them an enigma. If you balance a cold stone out, you are no longer balanced to the ground around it and you miss targets. If you try to discriminate them (with your ears) you'll miss nuggets. And because they are non-metallic, they are out of the realm of detectors discrimination abilities. Oh well, I have to dig them and so do you!

Depth-testing your machine on known targets

Air testing is worthless. Targets don't float six inches above the coil in real life.

That's why I built my own test gardens — and why I wrote the full guide for GPAA members. You want real-world depth and response testing in the exact soils that give you trouble.

Here's the quick version I still use:

- Collect five-gallon buckets of the nasty stuff from the field: black sand, hot rocks, cold stones, iron-rich soil.
- Use lead tire weights (hammered or melted into nugget shapes — never real gold) for targets: ¼-gram, ½-gram, 1-gram, 5-gram, ½-ounce, even 1-ounce for PI machines.
- Bury them at real depths: a ½-ounce target at 8-12-14 inches (drop 20%+ for VLF machines depending on coil and frequency).
- Space targets 3+ feet apart, non-linear layout, in buckets modified with drainage holes so the soil matures like real

ground.
- Water it after building, wait months for oxidation and settling, then test coils, settings, frequencies, and your own ears.

My current test garden is over 30 years old in 2026 and is about as close to naturally as possible.

I still visit my garden when I get a new machine or before a big trip. It keeps me honest and tells me exactly what I can expect in the field.

Hunting after rain, in snow, at night — when it's worth it

Record rain and snow? That's prime time. I've said it for years and the 2023 western gold rush proved it again — big moisture moves gold, exposes bedrock, and resets the patches. Hit GPAA claims right after a good soaker and you'll find stuff that was buried for decades.

Snow? Same deal in the right spots. Fresh snowmelt concentrates gold in the low spots. Just dress warm and watch for slippery rocks.

Night hunting? I've done it with a good headlamp when the signals are popping and the crowds are gone. Cooler temps, less EMI sometimes, and you can hear those faint whispers better with no distractions. Just make sure you have solid pinpointing and fill every hole — nobody wants to trip in the dark.

The ½-ounce nugget story and other finds that still make me smile

One of my favorite days started in the test garden. I had a ½-ounce lead target buried at 12 inches in hot, mineralized soil. Cranked the right settings, kept the coil glued to the ground, and it sang clear as day. Same day out on a real GPAA claim I hit a similar-sized real picker in ground that had been pounded for years. Same machine, same technique — the garden had prepped me perfectly.

Another time tracking black sand with the GMT I followed the streak right into a little pocket that produced a handful of nice pieces while the guys random-walking fifty yards away found nothing. Still smile every time I tell that one.

The best part? Every one of those finds started with the fundamentals we've covered in this book. The advanced stuff just makes the good days happen more often.

Action Steps — Do This to Level Up

1. Build (or start) your own test garden this weekend using the bucket method above. Include at least one ½-ounce-sized target in your worst local soil.
2. Spend one session depth-testing your current machine and every coil you own — log the results.
3. Next big rain or snow event, check the GPAA Online Mining Guide reports the next morning and hit a claim within 24–48 hours.
4. Try one night hunt on a familiar beach or park (with permission if needed) — headlamp, headphones, slow grid. Compare your finds.
5. Write down your own "war story" after every hunt — even the slow ones teach you something.

The curveballs never stop coming.

But once you've beaten them a few hundred times, they stop feeling like curveballs and start feeling like just another day in the dirt.

And those are the days that still make me smile after all these years.

Chapter 13: Detector Maintenance, Upgrades & Choosing Your Next Machine

We've tuned it, swung it, gridded it, and dug it. Now treat your detector like the faithful partner it is — because a well-cared-for machine will outlast every trend and still be finding gold when the next shiny new model comes out.

I've been swinging detectors since I built my first one at nine years old. I've owned and tested more brands than most people have coils. And after fifty-five years the one thing I know for sure is this: **maintenance isn't optional — it's how you keep the thing working when you need it most.**

Daily care, coil covers, batteries & storage — the stuff that actually matters

Post-hunt or after a training session, wipe down your detector shaft and coils with a damp cloth or a dust rag (for non-waterproof units). Let everything dry completely before you put it away. That simple habit keeps corrosion away and makes sure it powers up next spring without a hiccup.

Always use a coil cover. Period. I say it in every class: **Coil Covers are cheap. Missing gold nuggets isn't.**

If you're doing it right — coil sliding on the ground, low and slow — you're going to get debris in that cover. After a few hours it's common to find 80-mesh and finer stuff packed in there, along with black sand that can cause falsing. Clean it out periodically during the hunt and always at the end of the day. My SDC cover was brand new a week ago — now it looks like it's been through a war. That's exactly where I want it living: on

the ground, getting beat up so the coil stays protected. DO NOT USE Black electrical tape on your coils PLEASE! It gets hot and nasty and holds on to magnetite like crazy. If I use anything I use easy pull off white medical tape. No residue is the main reason.

Batteries? Take them out of the control housing before long-term storage. Some manufacturers recommend it to keep them from leaking or slowly draining. Trust me — you don't want to pull your machine out for the first hunt of the season and find a dead, corroded mess. If you get out a lot like I do, I pull my batteries everyday and charge them.

Store it in a cover, in a cool dry place, and you'll never wonder if it's going to power up when you need it.

When (and why) to upgrade — and what actually matters

Here's the truth I give every student who asks about buying a new machine: **master the fundamentals on whatever you have right now before you chase the next shiny box.**

The similarities between detectors are way bigger than the differences. Once you understand ripples, threshold, ground balance, sensitivity, coil control, and gridding, you can pick up almost any machine and hunt effectively.

Upgrade when:

- You've worn out the current one through real use (not just sitting in the closet).
- You're hunting specific ground that your current machine struggles with (super hot mineralization, deep large nuggets, tiny pickers in trash).
- You want a feature that genuinely saves time or increases finds in your terrain (better tracking, faster recovery, waterproofing for beaches).

But nine times out of ten, the biggest performance jump comes from **accessory coils**, not a whole new detector. That's why I always tell folks: buy the best your budget allows for what you actually want to do, then add coils as you learn the ground.

Coil selection guide (size, shape, DD vs concentric)

Coils are the easiest, cheapest way to change how your machine behaves in the field.

- **Size**: Small coil (5–6") for tiny gold, trashy areas, tight spaces. Large coil (11–18") for depth in milder to hot ground (not extreme) or covering big patches faster. Middle-of-the-road 8" or 10" is an everyday go-to for most detectorists — it balances sensitivity and coverage for most nugget and relic work.
- **Shape**: Round coils give the most even field. Elliptical are easier to swing in brush or between rocks. Open coils are lighter; and most are waterproof. Closed (Epoxy filled) are mostly waterproof and tougher in mud or water.
- **DD vs Concentric (or Mono on PI machines)**: DD coils handle mineralized ground better and have less "blade" effect in some soils. Concentric (or Mono) often give a stronger, more defined signal on small targets and in milder ground. I wasn't the biggest fan of DD coils when I started, but I've learned they have their place — especially in hot dirt. Test both in your ground. The right coil in the right scenario will outperform a more expensive machine with the wrong coil every single time. The biggest coil question that I get asked is when do I go to a DD coil. The answer is simple: When I can no longer make a concentric (VLF) or a Mono (PI) work by adjusting settings.
- **Tip** — If you are buying a VLF detector, check with the seller

or the manufacturer to make sure that you have options for a Concentric and DD coil availability. Some detectors are DD only and they were designed to maximize responses. If DD is the only option, make sure that it is because the detector was designed only and around that coil. Perfect example is the Minelab Gold Monster 2000.

Brand similarities > differences — the core message

At the end of the day, every detector sends ripples and listens for the return or has a conversation with the ground. Master the five fundamentals we've covered in this book and the brand name on the box matters a whole lot less. I've pulled gold with $200 starters and with flagship machines — the difference was the operator, not the logo.

Action Steps — Do This Today and Every Season

1. After your next hunt: wipe down the entire machine, clean the coil cover inside and out, remove the batteries if storing long-term.
2. Inspect your current coils — order a fresh skid plate or cover if it's worn.
3. Make a list of the ground you hunt most (hot mineralized, mild beach, trashy parks). Research 2–3 accessory coils that match those conditions.
4. Test one new coil in your test garden or on a familiar claim — log the difference in finds or stability.
5. Only after you've maxed out your current setup with coils and skills, start shopping for a new machine. Ask: "What specific problem will this solve for the ground I actually hunt?"

Take care of what you have.

Add the right coils at the right time.

Only upgrade when the machine is truly holding you back.

Your detector isn't a fashion statement — it's a tool.

Treat it right and it will keep bringing treasure home for decades.

Chapter 14: Your Path from Beginner to Expert — The 30-Day Plan

You now have every fundamental I've taught in fifty-five years of swinging a detector.

The only thing left is to put it all together — in the field, week after week, until it becomes muscle memory and you stop thinking about settings and start hunting like the expert you're about to become.

This is the exact 30-day plan I give every student who attends my Metal Detecting 101 sessions or joins me on a Gold Trails trip. It works. I've watched beginners go from "I don't know what any of these buttons do" to pulling their first real nugget or relic in under a month when they follow it.

Week 1: Fundamentals in Your Backyard (Days 1–7)

Build the foundation where it's easy and repeatable.

- Every day: 20–30 minutes of swing practice — low and slow, coil on the ground, 50% overlap, perfect threshold hum.
- Days 1–3: Factory preset → threshold → ground balance → sensitivity stable. Bury 3 test targets (coin, small lead "nugget," nail) at 4–6 inches. Swing until you can call the repeatable signal before you dig.
- Days 4–7: Add coil control drills — straight-line, eyes closed listening only. Log every signal in a notebook or phone note: tone, depth, what it was.
- End of week goal: Your swing is smooth, your ears hear the whispers, and the machine feels like an extension of your arm.

Week 2: Test Garden & Real-Ground Mastery (Days 8–14)

Take it from backyard to real dirt.

- Build or use your test garden (black sand, hot rocks, different gold sizes).
- Every session: Full setup (threshold, ground balance, sensitivity) in the exact soil you'll hunt most. Test every frequency/mode/coil you own.
- Daily drill: Depth-test one target at increasing depths. Note what works and what doesn't. Practice two-scoop recovery and filling holes perfectly.
- End of week goal: You know exactly what your machine can and cannot do in your local ground. The "complicated" feeling is gone.

Week 3: Grid Patterns on a Real Claim or Permission Spot (Days 15–21)

Time to hunt like you mean it.

- Pick one GPAA claim, public spot, or private permission area.
- Every hunt: Find or create one small hot zone and grid it completely — north-south, then east-west, southwest - northeast. Mark with rocks if you are on a claim or public area.
- Dig every signal. Log everything. Expand the grid every time you find something good.
- Add one advanced tip per session: after-rain hunt, night hunt, or black-sand tracking if your ground has it.
- End of week goal: You've covered ground systematically and found more in one small box than most people find wandering all day.

Week 4: Full Hunts, Logging, and Community (Days 22–30)

Put it all together and lock it in.

- Three or four full field days — different terrains if possible (beach, park, desert, relic site).
- Use everything: research first (GPAA Online Mining Guide or old maps), proper setup, grid, dig clean, log every find.
- Join the GPAA forum, your local chapter, or a detecting Facebook group. Post one question or one find photo with your settings and what you learned.
- Help a beginner if you can. To me this is the most important thing you can do to make you better. The moment that you are confident in your abilities enough to answer a question to another detectorist, is the day that your self-confidence will soar. That is the day that you become unstoppable.
- End of week goal: You're no longer a beginner. You're hunting with confidence, logging data like a pro, and part of the community that keeps this hobby alive.

Tracking Your Progress and Building Muscle Memory

Keep a simple log every single day:

- Date & location
- Settings used (threshold number, GB, sensitivity)
- Number of signals / number of good finds
- One thing that improved
- One lesson learned

After 30 days flip back through it. You'll be shocked how far you've come. Muscle memory doesn't come from reading — it comes from swinging. The more consistent your practice, the more automatic it all becomes.

Joining the community (GPAA, clubs, forums)

You don't have to do this alone. The GPAA and LDMA were built for exactly this — claims, events, fellow members who love sharing spots and tips. Show up to a Metal Detecting 101 session when I'm in your area, join a local club, or just jump on the forum and say "I just finished Kevin's book — who's hunting this weekend?"

The friends you make will be the ones who tip you off to new claims, help you with a tricky machine, and celebrate your first real nugget with you.

Final pep talk: Get out there — the gold (and treasure) is waiting

Thirty days from now you will not be the same person who opened this book.

You'll hear the ripples, or be a part of the PI conversation.

You'll feel the threshold hum in your bones.

You'll grid a patch and know — without guessing — that there's more there.

The machine in your hand is no longer complicated.

It's a tool.

And you now know how to use it better than 90% of the people swinging one right now.

So get out there.

Fill your holes.

Respect the land.

Dig every iffy signal.

And when you pull that first real piece of gold or that old silver coin that's been waiting decades for you...

Smile.

Because you earned it the right way.

The gold is still out there.

The relics are still buried.

The rings are still lost in the sand.

They've been waiting for someone who actually knows what they're doing.

That someone is you.

Now go find them.

— Kevin Hoagland

Conclusion: The Detector in Your Hand Is Only the Beginning

Drop the rock in the pond one more time.

Watch the ripple shoot out, hit that target, and come straight back to you.

That's still all that's happening under your coil.

We've taken what felt like a complicated control box with too many knobs and turned it into a simple, reliable tool that speaks the same language on every machine ever made.

You now know the fundamentals that 90% of detectorists never bother to learn.

You know how to listen to the threshold whisper instead of waiting for a scream.

You know how to make the earth disappear with a proper ground balance.

You know how to keep that coil sliding on the dirt, grid like you mean it, and dig every iffy signal.

Most importantly, you know it's **not complicated**.

I built my first detector at nine years old that started out a very high-end kit. When I saw where it lacked, I started over with a few new parts and with parts from a junk box sitting in the corner of a Radio Shack and a dream. Fifty-five years later I still get that same rush every time a good signal breaks through the threshold and I pull something out of the ground that hasn't seen daylight in decades, if ever. I still get the nugget shakes and I still sit and have a conversation with every nugget I find. "Hey there, let me take a good look at you and figure out where you came from. I want to go find your brothers and sisters."

That thrill never gets old.

The detector in your hand right now is only the beginning.

The real adventure starts the moment you walk out your door, hit a GPAA claim, grid a patch, and start filling your pockets instead of just swinging.

So get out there.

Join the GPAA or LDMA if you haven't already — the claims, the events, the people who love this as much as you do are waiting.

Build that test garden.

Wear out a coil cover.

Fill every hole.

Log every find.

And when you pull that first real nugget, that old silver coin, or that lost ring that's been waiting just for you...

Take a second. Smile. Take the picture. Then email it to me or post it on the forum so we can all celebrate with you.

Because that's what this hobby, sport and my lifestyle is really about.

The gold is still out there.

The relics are still buried.

The treasure is still waiting.

And now you've got everything you need to go find it.

See you on the claims, my friend.

Get after it.

— Kevin Hoagland

Executive Director of Development, GPAA/LDMA

Host, *Gold Trails*

"I'll meet you out there somewhere someday on a GPAA claim."

P.S.

One last thing that I want to talk about. And that is Zen and the art of Detecting.

This is going to be very blunt and very to the point.

If your head is not in the game when you are out prospecting for gold or in a civil war battlefield looking for relics, go home and come back another day.

I've watched this happen more times than I would like to even think about and it's happened to me at times as well. I'm heading out for a day of detecting and then I start wondering if I closed the garage door or turned off the coffee pot. Or even worse thinking about something that's not right in your life maybe it's a bill that's due. It doesn't matter what it is but if you let it rent space in your brain there's a good chance you're going to fail that day.

Earlier I talked about the perfect time. That time when you are matched to your machine and the machine is matched to the ground and no matter what you can't fail except that's not entirely true. You can fail and 99% of the failure comes from letting the outside come into your head while you're detecting.

I'm a huge fan of the poet Blake and there are times when things will creep into my head that preoccupy my detecting time when that happens, I just stop, take a moment clear my head and the next thing I know my inner voice is starting to recite, tiger tiger burning bright. That generally gets me out of my own way to where I see, hear and feel everything that the ground is telling me.

See nugget hunting or coin and relic hunting is more than a way for me to just get out and do something it's my lifestyle. And I never go out detecting thinking that I have got to find something. I never tell myself I have to find the piece of gold or I have to find a civil war breastplate.

Each time I go out I know that I have the tools and the experience to find gold or a relic or a coin (same as you do now if you'll follow some of my steps that I've listed for you) I never set myself up for failure by saying I have to find something amazing and not.

My absolute best days of detecting is when I just go out set my machine to the ground set the machine to me and go hunt. No expectations, nothing rattling around in my head, just the sound of the threshold in my headphones and the coil scraping the ground. When that's all you hear you'll be surprised at the targets you'll find.

Okay, now you can go out, learn your detector, listen to everything and find good stuff buried either since the beginning of this planet or when people have been. Don't count on luck, because LUCK FAVORS THE PREPARED.